Antioxidant status in humans after consumption of blackberry (Rubus fruticosus L.) juices with and without defatted milk.

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The present study was designed to evaluate the possible effect of the consumption of blackberry juices (BJ) prepared with water (BJW) and defatted milk (BJM) on the plasma antioxidant capacity and the enzymatic and nonenzymatic antioxidants. A significant ($p < 0.05$) increase in the ascorbic acid content in the plasma was observed after intake of both BJs. However, no changes were observed in the plasma urate and alpha-tocopherol levels. An increase on the plasma antioxidant capacity, by ORAC assay, was observed only after consumption of BJW but not statistically significant. Plasma antioxidant capacity had a good positive correlation with ascorbic acid ($r = 0.93$) and a negative correlation with urate level ($r = -0.79$). No correlation was observed between antioxidant capacity and total cyanidin or total ellagic acid contents. Further, it was observed that plasma catalase increased following intake of BJ's. No change was observed on the plasma and erythrocyte CAT and glutathione peroxidase activities. A significant decrease ($p < 0.05$) in the urinary antioxidant capacity between 1 and 4 h after intake of both BJs was observed. A good correlation was observed between total antioxidant capacity and urate and total cyanidin levels. These results suggested association between anthocyanin levels and CAT and a good correlation between antioxidant capacity and ascorbic acid in the human plasma after intake of BJs. Follow-up studies investigating the antioxidant properties and health benefits are necessary to demonstrate the health benefits of polyphenols.